

ChinookAnalytics: Final Analytical Report (Marts)

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Abstract

This report answers the analytical objectives defined in *Analysis Objectives*, using only the curated `marts` layer. All numbers below are derived from `mart` outputs produced by the SQL pipeline.

1 Executive Summary

- **Total revenue:** 2328.60
- **Total invoices:** 412
- **Customers:** 59 **Billing countries:** 24
- **Concentration snapshot:**
 - Top-1 artist share: 5.95%
 - Top-5 artists share: 21.57%
 - Top-10 artists share: 30.98%
 - Top-1 country share: 22.46%
 - Top-5 countries share: 58.78%
 - Top-10 countries share: 76.05%
 - Top-10 customers share: 19.38%
 - Top-20 customers share: 36.61%

2 Objective 1: Revenue Health Over Time

Questions answered: revenue evolution monthly/yearly, growth behavior, invoice value trend, volatility signals.

Key Results

- Revenue and invoice volume are reported at a month grain (60 months).
- Total revenue equals 2328.60 across all months (validated).
- *Best/worst months and strongest MoM/YoY changes are available in `marts.mart_revenue_time`.*

Required Output Tables

- `marts.mart_revenue_time`: month_date, invoices, revenue, avg_invoice_value, MoM, YoY.
- Yearly roll-up computed from monthly mart.

Actionable Interpretation (from marts)

- Use MoM and YoY columns to identify periods of acceleration/slowdown.
- Use revenue standard deviation by year (derived from the monthly mart) to quantify volatility.

Note: Insert 2–3 concrete highlights after extracting from marts (e.g., best month, worst month, peak growth month).

3 Objective 2: Market Performance and Geographic Strategy

Questions answered: which markets drive revenue, market value per customer, and geographic concentration.

Top Markets by Revenue (from marts)

- USA: 523.06 (Top-1 country share = 22.46%)
- Canada: 303.96
- France: 195.10
- Brazil: 190.10
- Germany: 156.48

Market Concentration (from marts)

- Top-5 countries represent 58.78% of total revenue.
- Top-10 countries represent 76.05% of total revenue.

Interpretation

- Revenue is moderately concentrated geographically: a small set of markets drives the majority of revenue.
- `marts.mart_market_performance` provides revenue per customer by country to compare market value beyond volume.

4 Objective 3: Customer Economics and Revenue Concentration

Questions answered: customer LTV ranking, distribution shape, and Top-N dependency.

Customer Base Summary (from marts)

- Customers: 59
- Invoices per customer: min 6, max 7, average 6.98
- LTV range: 36.64 to 49.62
- Average LTV: 39.47 Median LTV: 37.62

Revenue Concentration (from marts)

- Top-10 customers represent 19.38% of total revenue.
- Top-20 customers represent 36.61% of total revenue.

Interpretation

- Customer revenue is not extremely concentrated: Top-20 customers account for 36.6% rather than a majority.
- The narrow invoice-count distribution (6–7 purchases each) indicates limited separation between “high frequency” and “low frequency” segments.

5 Objective 4: Catalog Strategy (Artists and Genres)

Questions answered: top artists/genres by revenue and units, and concentration risk in the catalog.

Top Artists by Revenue (from marts)

- Iron Maiden: 138.60 (140 units)
- U2: 105.93 (107 units)
- Metallica: 90.09 (91 units)
- Led Zeppelin: 86.13 (87 units)
- Lost: 81.59 (41 units)

Top Genres by Revenue (from marts)

- Rock: 826.65 (835 units)
- Latin: 382.14 (386 units)
- Metal: 261.36 (264 units)
- Alternative & Punk: 241.56 (244 units)
- TV Shows: 93.53 (47 units)

Catalog Concentration (from marts)

- Top-1 artist share: 5.95%
- Top-5 artists share: 21.57%
- Top-10 artists share: 30.98%

Interpretation

- The catalog is not dominated by a single artist, but Top-10 artists still represent 31% of revenue.
- Genre mix is more skewed: Rock alone represents a very large revenue block, indicating genre dependence.

6 Objective 5: Invoice Structure and Revenue Composition

Questions answered: what defines high-value invoices; whether value is driven by diversity/lines.

Invoice Total Distribution (from marts)

	P25	Median	P75	P90	P95
Invoice Total	1.98	3.96	8.91	13.86	13.86

Top-10 Highest Invoices (from marts)

Invoice	Customer	Date	Country	Total
404	6	2025-11-13	Czech Republic	25.86
299	26	2024-08-05	USA	23.86
96	45	2022-02-18	Hungary	21.86
194	46	2023-04-28	Ireland	21.86
201	25	2023-05-29	USA	18.86
89	7	2022-01-18	Austria	18.86
88	57	2022-01-13	Chile	17.91
306	5	2024-09-05	Czech Republic	16.86
313	43	2024-10-06	France	16.86
208	4	2023-06-29	Norway	15.86

Interpretation

- High-value invoices are far above the median (Median = 3.96 vs Max = 25.86).
- `marts.mart_revenue_composition` provides invoice-level decomposition fields (lines, distinct artists/genres/media types) to explain whether high totals are driven by diversity vs invoice size.

7 Objective 6: Dependency and Concentration Snapshot

Questions answered: dependency on top artists/countries/customers.

Snapshot Results (from marts)

Snapshot date: 2026-02-13

- Total revenue: 2328.60
- Artists: Top-1 = 5.95%, Top-5 = 21.57%, Top-10 = 30.98%
- Countries: Top-1 = 22.46%, Top-5 = 58.78%, Top-10 = 76.05%
- Customers: Top-10 = 19.38%, Top-20 = 36.61%

Interpretation

- Country dependency is stronger than customer dependency (Top-10 countries = 76%).
- Artist dependency is moderate (Top-10 artists = 31%).
- This indicates resilience against single-artist risk, but meaningful exposure to market concentration.

8 Conclusion

Across time, market, customer, and catalog lenses, the marts layer provides a consistent and validated view of:

- where revenue is generated,
- how it is distributed across customers, markets, and catalog dimensions,
- and where concentration risk exists.